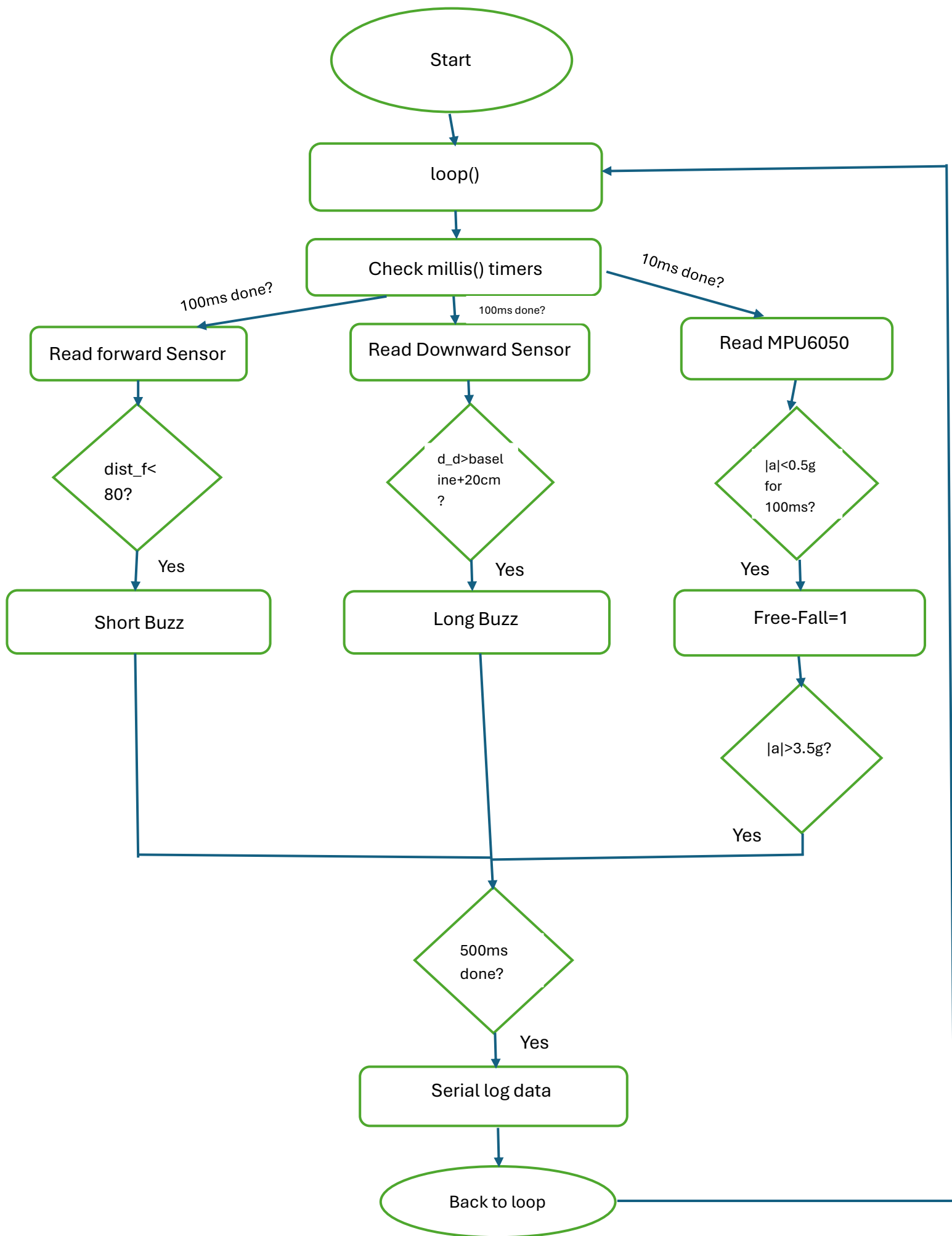




Notations:

- dist_f = Forward Obstacle Distance (cm)
- d_d = Drop Distance (cm)
- $|a|$ = MPU6050 Magnitude (g)

5-line answer: In your walking stick, you used indoor and outdoor modes with different thresholds. How would you implement adaptive thresholds in firmware — no manual mode switching?

- Adaptive thresholds can be implemented in firmware by monitoring the environmental sensor data continuously and calculating the moving average of normal readings. The firmware dynamically adjusts the obstacle detection limits based on the variations in the surroundings. So, in our case, outdoor mode could have larger distance fluctuations hence the threshold could increase automatically whereas in indoor mode we usually have small and stable readings, so threshold is lowered for better sensitivity. This removes the need for a manual switch for mode change.